



The mechanism by which thermal coupling leads to an asymptotic maldistribution stability boundary for flow boiling in parallel channels

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ABSTRACT

Phase change heat transfer processes are attractive for the efficient thermal management of advanced semiconductor devices, with flow boiling in parallel microchannel heat sinks being one promising solution. However, there are several implementation challenges associated with two-phase flow in parallel microchannels, particularly flow maldistribution, which can adversely affect performance and reliability. Two-phase flow models can be useful in predicting and understanding the instability mechanisms that leads to maldistribution, such that parametric performance trends and safe regions of operation can be identified. Channel-to-channel thermal coupling has been shown to alleviate maldistribution, but it is not yet understood why this effect is limited to some critical lateral thermal conductance above which there is no further benefit. In the current work, a two-phase flow model with a lumped thermal capacitance representation of the heat sink wall is developed to identify the mechanism for this asymptotic maldistribution stability boundary for thermally coupled microchannels. The lumped model allows the nature of stability boundary to be explained via a scaling analysis of the eigenvalues. In such systems, it is identified that a single eigenvalue determines the occurrence of flow maldistribution. Scaling analysis shows that, for small values of thermal conductance, this eigenvalue becomes a quadratic function of thermal coupling, which enhances the stability of microchannels and moves the stability boundary. However, for large values of thermal conductance, the eigenvalue is dependent only on the fluid properties, leading to a limit at which thermal coupling can no further improve stability or affect the stability boundary. The lumped model developed in this work enables mechanistic understanding of the role of channel-to-channel thermal coupling on mitigating flow maldistribution and therefore may offer important guidance on the design of flow boiling systems and heat sinks.

1. Introduction

Two-phase flow heat exchange processes are used in many industrial applications such as steam generation, air conditioning, and nuclear reactor cooling [1,2]. Flow boiling using parallel microchannel heat sinks is one such thermal management application that aims to enhance the reliability and performance of computing clusters, radar technologies, power conversion systems, and electric drive motors by efficiently dissipating intense, localized heat fluxes. Compared to single-phase operation, these two-phase heat sinks feature higher heat transfer coefficients, lower pumping power requirements, and more compact designs. However, one practical challenge is the occurrence of flow maldistribution and instabilities that can degrade their heat transfer performance compared to uniform flow and stable operation [3].

In practical applications, microchannel heat sinks have many

parallel channels (versus a single channel) to provide high heat transfer surface areas. However, this renders the heat sink further susceptible to flow maldistribution between these many parallel channels, triggered by Ledinegg instability or other flow instabilities (e.g., density wave oscillations, pressure drop oscillations, and temperature oscillations) underlying flow instability mechanisms [4–9]. Flow maldistribution, as it pertains to the focus of this work, refers to a steady flow distribution that is non-uniform from channel to channel.

1.1. The mechanism of steady two-phase flow maldistribution

For a fixed heat input q' to a single microchannel with subcooled inlet temperature T_{in} , the pressure drop ΔP demand curve as a function of mass flux G_{ch} is a non-monotonic N-shaped curve (e.g., the solid red line or dashed green line shown in Fig. 1(a)). This N-shaped demand curve consists of three regions, a single-phase vapor outlet region where

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Nomenclature			
A_c	channel cross-section area (m ²)	q'	heat transferred to fluid per unit length (W/m)
A_s	heat transfer area (m ²), ($4W_cL_c$)	q'_0	heat input to channel outer surface per unit length (W/m)
C	Chisholm parameter	T_f	Fluid temperature (°C)
C_{lat}	lateral thermal conductance per unit channel length (W/mK)	T_{in}	inlet temperature (°C)
Co	confinement number, $[g(\rho_f - \rho_g)D_h^2/\sigma]^{-0.5}$	T_{sat}	saturation temperature (°C)
$c_{p,\phi}$	vapor/liquid specific heat (J/kgK), $\phi = f, g$	$T_{w,i}$	wall temperature (°C), $i = 1, 2$
c_w	heat sink wall specific heat (J/kgK)	$T_{w,i}^*$	equilibrium wall temperature (°C)
D_h	hydraulic diameter (m)	t	time (s)
F_F	frictional pressure gradient	u	velocity (m/s)
F_G	gravitational pressure gradient	W_c	square channel width (m)
f_ϕ	friction factor ($2\frac{\tau_{w,\phi}}{G_{in}^2}$), $\phi = f, g$	W_{hs}	square heat sink wall width (m)
g	acceleration due to gravity (N/m ²)	x	vapor quality ($\frac{h-h_f}{h_g-h_f}$)
$G_{ch,i}$	mass flux to channel (kg/m ² s), $i = 1, 2$	z	coordinate along channel length (m)
$G_{ch,i}^*$	equilibrium mass flux to channel (kg/m ² s)	$\Delta P_{ch,i}$	channel pressure drop (Pa) ($P_{in} - P_{atm}$), $i = 1, 2$
G_{in}	total mass flux to channels (kg/m ² s)	ΔP_A	acceleration pressure drop (Pa)
$h_{conv,tp}$	two-phase flow heat transfer coefficient (W/m ² K)	ΔP_F	frictional pressure drop (Pa)
$h_{conv,\phi}$	vapor/liquid heat transfer coefficient (W/m ² K), $\phi = f, g$	$\delta G_{ch,1}^*$	perturbation of mass flux (kg/m ² s)
$h_{fb,i}$	flow boiling heat transfer coefficient (W/m ² K)	$\delta T_{w,i}^*$	perturbation of wall temperature (°C)
h_{nb}	nucleate boiling heat transfer coefficient (W/m ² K)	$\left(\frac{\partial p}{\partial z}\right)_f$	liquid frictional pressure gradient ($2f_f\frac{v_f(1-x)^2G^2}{D_h}$)
h	enthalpy (kJ/kg)	$\left(\frac{\partial p}{\partial z}\right)_g$	vapor frictional pressure gradient ($2f_g\frac{v_gx^2G^2}{D_h}$)
h_{in}	inlet enthalpy of liquid (kJ/kg)	Subscripts	
h_ϕ	liquid/vapor saturation enthalpy(kJ/kg), $\phi = f, g$	A	acceleration
h_{fg}	Enthalpy of vaporization (kJ/kg)	F	friction
I	identity matrix	f	liquid
J	Jacobian Matrix	G	gravitation
k_ϕ	vapor/liquid thermal conductivity (W/mK), $\phi = f, g$	g	vapor
L_c	length of channel (m)	i, j	channel indices
m_w	mass of heat sink wall (kg)	Greek symbol	
M	molecular weight (kg/kmol)	α	void fraction
p	pressure (Pa)	β	aspect ratio of channel ($\frac{W_c}{H_c}$)
p_{cr}	critical pressure (Pa)	λ	eigenvalues of the Jacobian matrix
Pr_ϕ	vapor/liquid Prandtl number ($\frac{\mu_\phi c_{p,\phi}}{k_\phi}$), $\phi = f, g$	μ_ϕ	liquid/vapor dynamic viscosity (Pa·s), $\phi = f, g$
P_{atm}	atmospheric pressure (Pa)	ρ_f	liquid/vapor density (kg/m ³), $\phi = f, g$
P_{in}	channel inlet pressure (Pa)	ρ_w	heat sink wall density (kg/m ³)
$Re_{\phi 0}$	combined vapor and liquid Reynolds number, ($\frac{G_{ch,i}D_h}{\mu_\phi}$), $\phi = f, g$	σ	surface tension (N/m)
		τ_w	shear stress (Pa)

pressure drop increases with mass flux (Region I), a liquid-vapor mixture outlet region where pressure drop decreases with mass flux (Region II), and a single-phase liquid outlet region where the pressure drop again increases (Region III); further description of the two-phase flow behaviors that lead to the trends in this curve is provided in [4]. The stability of the operating point, P_0 , located in the liquid-vapor mixture region depends on the pump characteristics [8,10]. For a fixed mass flux positive displacement pump (vertical dashed black line), the flow is stable for a single microchannel. For a constant pressure pump supply (horizontal orange dashed line), the system has three possible states of operation, and per the Ledinegg instability [10], the negative-slope Region II with a liquid-vapor mixture at the outlet is unstable leading to a flow excursion to the stable points.

Extending the above discussion, there is also a pressure drop load curve for systems of multiple channels in parallel. For example, the inset plot within Fig. 1(a) shows the pressure drop load curve for two identical microchannel as a function of the total mass flux ($G_{in} = G_{ch,1} + G_{ch,2}$). Apart from the N -shaped curve (solid black line) in the intermediate mass flux region, there are additional possible operating points (solid light blue lines) that arise due to the possibility unequal flow

distributions. For parallel microchannels, one added constraint is that the pressure drop across each individual microchannel must be equal (i. e., $\Delta P_{ch,1} = \Delta P_{ch,2}$). This condition can hold true for equal mass flux through both channels ($G_{ch,1} = G_{ch,2}$), corresponding to the N -shaped black solid line in Fig. 1(a) (segments A-B, B-E, and E-F). For example, Fig. 1(b) shows the flow distribution at operating point, P_0 , where both parallel channels have the same two-phase exit quality. Alternatively, the pressure drop constraint can also hold true unequal mass flux through two parallel microchannels ($G_{ch,1} \neq G_{ch,2}$), corresponding to the solid light blue lines in in Fig. 1(a). Three possible combinations of unequal mass flow may exist for a system of two parallel microchannels, corresponding to each light blue segment. Namely, (i) the first channel at a very low mass flux and the second channel at an intermediate mass flux (segment B-C), (ii) the first at a very low mass flux and the second at a very high mass flux (segment C-D), and (iii) the first at an intermediate mass flux and the second at a very high mass flux (segment D-E).

For the parallel channel system, the two-phase operating point P_0 with a desired uniform mass flux through channels, $G_{ch,1} = G_{ch,2} = G_{in}/2$, is subject to a constant pressure drop constraint ($\Delta P = \Delta P_{ch,1} = \Delta P_{ch,2}$) regardless of the pump characteristics. Hence the point is un-

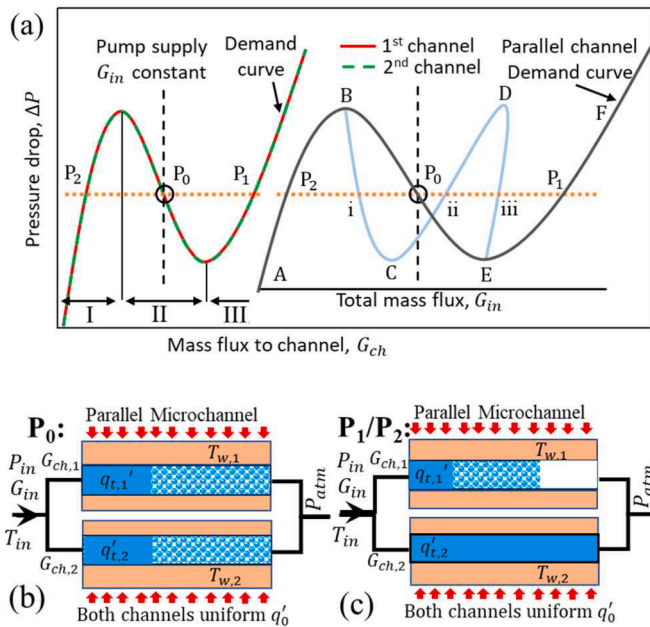


Fig. 1. (a) Pressure drop demand curve as a function of mass flux for flow through two identical microchannels subject to uniform heating with a subcooled liquid inlet. The demand curve for each identical channel (solid red and green dashed lines) has three regions, Regions I, II, and III, based on channel exit quality. For two parallel channels, a demand curve has regions with equal flow through both channels (solid black lines) or maldistribution (solid light blue lines). (b) If point operating point P_0 were stable, then both microchannels would have the same mass flux and two-phase exit vapor quality; (c) however, because P_0 is unstable due to the Ledinegg instability, the flow becomes maldistributed and one microchannel has a lower mass flux (P_2 with vapor at the outlet) while the other has a higher mass flux (P_1 with a subcooled liquid outlet). In the illustrations above, it is assumed that the channels are thermally isolated from each other.

stable due to the Ledinegg instability, and the system will settle to a stable operating point with flow maldistribution. In the case of flow maldistribution, one channel may have such a low mass flux that all of the incoming liquid is evaporated and the exit becomes single-phase vapor, while the second channel has such a high mass flux there the liquid remains subcooled up to the channel exit, as illustrated in Fig. 1(c). Comparing the uniformly distributed flow in Fig. 1(b) to the maldistributed flow in Fig. 1(c), it is expected that the heat transfer coefficient would be deteriorated due to flooding of one channel with liquid; the flow-starved microchannel with single-phase vapor at the outlet may also potentially initiate a premature critical heat flux (CHF).

1.2. Literature review

Although two-phase microchannel heat sinks provide very high heat transfer rates, many experimental works [9,11–15] have observed flow maldistribution owing to the Ledinegg instability. Maldistribution causes some channels to be starved of flow relative to a uniform distribution, and premature critical heat flux (CHF) can be triggered [16–18]. As a result, a number of control strategies have been explored to mitigate flow maldistribution, such as inlet restrictors [13,18,19], increased system pressure [20], and dynamic control of pumps and valves [21,22]. Nevertheless, these solutions either increase system complexity or lead to increased pressure drop. Hence, it becomes crucial to investigate the behavior of two-phase flow maldistribution in parallel channels and to formulate accurate models that aid in predicting and controlling the flow in two-phase heat sinks.

A number of studies have experimentally characterized the flow maldistribution caused by the Ledinegg instability [9,11,14,15,23]. The

notable, pioneering work of Akagawa et al. [23] investigated flow maldistribution in very long parallel evaporator tubes with millimeter-scale diameters. They experimentally obtained channel pressure drop curves for individual tubes to reveal the N-shaped load curve as illustrated in Fig. 1(a). The flow distribution, excursion, and hysteresis phenomena were also observed in parallel tube experiments. A stability criterion was derived from a Laplace transformation of the linearized momentum and continuity equations for the system of parallel tubes, which explained the significant deviations from the uniform flow distribution when any one of the channel load curves had a negative slope. Miglani et al. [11] experimentally investigated the flow distribution and wall temperature in two isolated parallel microchannels. In their study, the effect of heat input on these quantities was measured. Flow distribution and wall temperature were found to be uniform at low heat input as with single-phase flow at subcooled conditions. Whereas at relatively higher heat input, once boiling started inside the channels, the flow distribution became unstable because of Ledinegg instability, which induced flow maldistribution in the channels. The wall temperature in the starved-flow channels increased due to deteriorated heat transfer performance in that channel. In another work, Szczukiewicz et al. [13] performed experiments for visualization flow distribution and temperature in many parallel microchannels. In their study, the microchannels were prone to flow instabilities, backflow, and flow maldistribution without any inlet restrictors, leading to high amplitude and high-frequency temperature and pressure oscillations. The recent work of Clark et al. [24] experimentally analyzed flow distribution in the parallel microchannel heat sink. In their work, some channels were repeatedly found to be flow starved, confirmed by high-speed flow visualization. During this condition, the average heat sink temperature was higher than with uniform flow through the channels.

Minzer et al. [25] investigated flow distribution in two parallel channels with a simplified two-phase flow model where the channel pressure drop was calculated for flow boiling in each single channel. The governing model equations were used to obtain steady-state solutions for the flow distribution and inlet pressure of the channels. The transient equations were used to perform a linear stability analysis and predict the flow dynamics in parallel channels. Baikin et al. [26] extended the model of Minzer et al. [25] to four parallel channels and performed experiments with various combinations of heated and unheated channels. Flow maldistribution was observed even for the case of equal heating, confirmed via linear stability analysis, and transient simulations verified the runaway trajectories from unstable operating points. Recently, Van Oevelen et al. [27] studied flow distribution for a large number of identical parallel microchannels (up to 200) as is commonly found in heat sinks. A generalized linear stability analysis was performed to show that, even with a constant mass supply pump, the stability behavior of the system having many parallel microchannels was prone to flow maldistribution, and converged toward that system of constant pressure supply pump with increasing number of channels.

The works discussed above all investigated the flow maldistribution for thermally isolated parallel channels, that is, without incorporating lateral thermal coupling (i.e., lateral heat transfer) between parallel channels. However, there may be a high lateral thermal conductance between channels manufactured within the same heat sink material, which plays a critical role in mitigating the flow maldistribution caused by the Ledinegg instability. Flynn et al. [28,29] experimentally measured a difference in temperature between two parallel microchannels etched into a substrate depending on whether the channels were either thermally coupled or thermally isolated, and attributed their observations to the effect of thermal coupling on the Ledinegg instability. In their studies, flow visualization confirmed the severe flow maldistribution in thermally isolated channels compared to thermally coupled ones. Due to maldistribution, the temperature difference between flow-starved and flooded channels was experimentally reported to be 40°C, whereas thermally, it was less than 10°C. Van Oevelen et al. [30] were first to model this effect of lateral thermal coupling on flow

distribution in two parallel microchannels by extending their model from Ref. [27]. A finite volume discretization method was used to solve the model equations. The severity of flow maldistribution was compared for thermally isolated and thermally coupled parallel channels. It was found that the severity of flow maldistribution was significantly reduced for thermally coupled parallel channels because a channel experiencing local dryout could transfer some heat input to the neighboring channel, maintaining a more uniform wall temperature. Miglani et al. [9] confirmed this behavior by experimentally measuring the flow maldistribution in two parallel channels and showing that the mass flux difference between the parallel channels was significantly reduced for thermally coupled parallel channels. A direct comparison between the experimental data and theoretical model predictions [30] showed good agreement. The model [30] was also validated against the experimental data of Flynn et al. [28,29], showing reasonable agreement.

The above results demonstrate that the stability behavior of microchannels can be improved by allowing lateral heat transfer between the channels, which reduces the flow maldistribution severity. However, an important but unexplained observation from Ref. [30] is that the lateral thermal coupling is not guaranteed to avoid flow maldistribution. At low heat inputs, increasing the lateral thermal conductance shifts the stability boundary to the point where maldistribution can be avoided altogether; however, the effect of increasing thermal conductance becomes weaker and weaker as the applied heat input increases. Above some upper heat input threshold, the system is unstable and the microchannels are prone to maldistribution no matter how high the lateral thermal conductance. There is no available mechanistic explanation for this asymptotic maldistribution stability boundary because the previous studies have used finite volume discretization solutions that do not permit physical interpretation of the system stability. Therefore, this current work develops a dynamic model for predicting and analyzing the flow distribution in parallel microchannels using a lumped capacitance representation of a heat sink that allows mechanism explanation of the effect of thermal coupling on the stability boundary. The reduced-order model is first shown to capture the correct trends in flow distribution, that is, the stability of the parallel microchannels increases with lateral thermal coupling and reaches a limit at higher heat inputs. A single eigenvalue is shown to be the cause for flow maldistribution for two parallel channels. Scaling analysis inspection of the terms in this eigenvalue then offers a mechanistic explanation for the asymptotic stability boundary behavior.

2. Modeling approach

Fig. 2(a) illustrates the system having flow through two identical microchannels placed in parallel with a common inlet and outlet header. The adjacent channels are thermally coupled via a lateral thermal conductance, C_{lat} , representative of conduction between the channels through the heat sink wall. The system reduces to the case of thermally isolated channels when $C_{lat} = 0$. The channels are subject to constant total mass flux G_{in} from a positive displacement pump at the inlet

header. The mass flow splits and each microchannel receives some portion of the inlet mass flow; e.g., for uniformly distributed flow, $G_{ch,1}^* = G_{ch,2}^* = G_{in}/2$ at steady state conditions. Both channels also receive a uniform, equal heat input per unit channel length q'_0 on their outer surfaces. Subcooled liquid enters the channels at T_{in} and fluid exits as a subcooled liquid, liquid-vapor mixture, or superheated vapor, depending on the heat input per unit length and mass flux. While the conclusions drawn in this study are qualitatively generalizable, quantitative results are shown for baseline microchannel dimensions and materials that are representative of applications for the thermal management of advanced semiconductor devices, with all model input parameters given in Table 1 and geometries defined in Fig. 2(b).

2.1. Dynamic flow distribution model

Transient governing equations are developed to analyze flow distribution in the parallel microchannel configuration defined by Fig. 2. The flow distribution model is adapted from our previous work on flow distribution in parallel channels [31]. A one-dimensional partial differential momentum equation is derived that describes the flow for the 1st microchannel as

$$\frac{\partial G_{ch,1}}{\partial t} + \frac{\partial(G_{ch,1}u)}{\partial z} = -\frac{\partial p}{\partial z} - F_F \quad (1)$$

where the first term on the left-hand side is rate of change of mass flux, the second term on the left-hand side is the acceleration pressure gradient, and F_F is frictional pressure gradient. A simplified lumped dynamic momentum equation for the heat sink is obtained by integrating Eq. (1) along the channel length from the inlet to the exit, and the new equation is given as:

$$\frac{dG_{ch,1}}{dt} = \frac{1}{L_c} (P_{in} - P_{atm} - \Delta P_F - \Delta P_A - \Delta P_G) \quad (2)$$

where L_c is the length of the microchannel, P_{in} and P_{atm} are inlet and

Table 1

Dimensions of the microchannel, system parameters, and operational parameters.

Parameter	Value
Channel length, L_c (mm)	10
Square channel width, w_c (μm)	200
Hydraulic diameter, D_h (μm)	200
Heat input per unit length, q'_0 (W/m)	220–250
Outlet pressure, P_{atm} (kPa)	101
Mass flux, G_{in} ($\text{kg}/\text{m}^2\text{s}$)	700
Fluid	Water
Square heat sink wall, W_{hs} (μm)	300
Inlet subcooling, $T_{sat} - T_{in}$ (K)	20
Heat sink wall density, ρ_w (kg/m^3)	2329
Wall specific heat capacity, c_w (J/kgK)	700
Lateral thermal conductance, C_{lat} (W/mK)	0–600

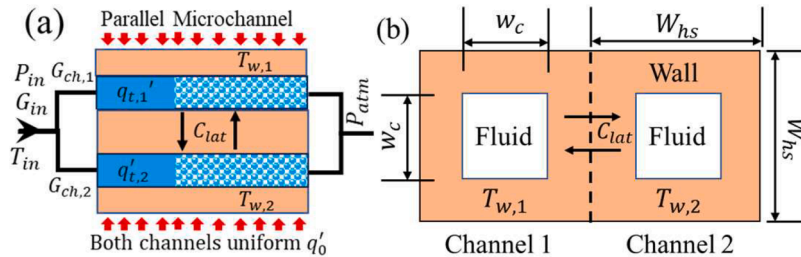


Fig. 2. (a) Schematic diagram of the scenario investigated to assess flow distribution in thermally coupled parallel microchannels; lateral heat transfer is possible when there is a temperature difference ($T_{w,1} \neq T_{w,2}$) between the channels depending on the magnitude of the lateral thermal conductance C_{lat} . (b) Cross-sectional view of the thermally coupled parallel microchannel with relevant dimensions defined.

outlet pressure of the microchannel, respectively, and ΔP_F , ΔP_A , and ΔP_G are the frictional, acceleration, and gravitational pressure drop, respectively. The total pressure drop of the channel is $\Delta P_{ch,1} = \Delta P_F + \Delta P_A + \Delta P_G$.

Mass conservation across both parallel microchannel systems for a constant total mass flux at the inlet G_{in} can be written as follows:

$$G_{in} = G_{ch,1} + G_{ch,2} \quad (3)$$

where $G_{ch,2}$ is the mass flux through the 2nd microchannel. The known constant total mass flux G_{in} and the mass flux $G_{ch,1}$ found by solving Eq. (2) provide enough information to obtain $G_{ch,2} = G_{in} - G_{ch,1}$ from Eq. (3). Note that the distinction between the 1st and 2nd channel is arbitrary, given the symmetry of the system.

The models used to calculate the microchannel pressure drop components ΔP_F , and ΔP_A with wall thermal capacity are detailed in our previous work [4,31]. Note that when a wall thermal capacity is introduced, the instantaneous heat input $\dot{q}$ to the fluid used for evaluating the pressure drop is not necessarily equal to the applied heat input $\dot{q}_0$ (i.e., $\dot{q} \neq \dot{q}_0$), which requires a separate dynamic thermal model, as is described in the next section.

2.2. Dynamic thermal model

Fig. 2 depicts the cross-section of the thermally coupled parallel microchannels receiving uniform heat input per unit channel length ($\dot{q}_0$) on the outer surface of the heat sink wall. The fluid temperatures in each channel are equal for a uniform fluid flow through both parallel microchannels, i.e., $T_{w,1} = T_{w,2}$. In contrast, any non-uniform flow distributions will result in a wall temperature difference ($T_{w,1} \neq T_{w,2}$), resulting in lateral heat transfer from the 1st to the 2nd microchannel or vice versa via lateral thermal conductance (C_{lat}). For flow maldistribution in a parallel microchannel heat sink with a finite thermal capacity, the fluid-to-wall temperature difference and convective heat exchange varies in a coupled manner, making the dynamic heat response more complex. These heat transfer processes are simultaneously considered in the modeling approach by considering the dynamic interaction between thermal wall capacity and fluid flow, in addition to lateral thermal coupling. Consequently, lumped thermal models for both the 1st and 2nd microchannel are developed by balancing the sensible energy change with the heat input to the wall, the heat transferred from the wall to the fluid, and lateral heat transfer between the channels, expressed for the system as

$$\frac{dT_{w,i}}{dt} = \frac{\dot{q}_0 L_c}{m_w c_w} - \frac{h_{fb,i} A_s (T_{w,i} - T_{f,i})}{m_w c_w} - \frac{C_{lat} L_c (T_{w,i} - T_{w,j})}{m_w c_w} \quad (4)$$

where, $i \neq j = 1, 2$ are the channel indices. The first term on the right-hand side of Eq. (4) is the total heat input to the heat sink wall, the second term on the right-hand side is the internal convection heat transfer to the fluid ($\dot{q}$), and the last term is the heat transfer due to thermal coupling between microchannels. The lateral thermal conductance, C_{lat} , defines the strength of the thermal coupling. Also, $m_w c_w$ is the wall thermal capacity, $h_{fb,i}$ is the flow boiling heat transfer coefficient of the i^{th} microchannel, and A_s is convective heat transfer area in the microchannel. The fluid temperature $T_{f,i}$ of i^{th} microchannel is calculated by taking the length-weighted-average temperatures of the subcooled liquid phase, vapor-liquid mixture, and superheated vapor phase along the flow length.

The internal convection heat transfer to the fluid $\dot{q} = h_{fb,i} A_s (T_{w,i} - T_{f,i})$, where $i = 1, 2$, is dependent on the flow boiling heat transfer coefficient ($h_{fb,i}$), calculated using the correlation from Bertsch et al. [32]:

$$h_{fb} = h_{nb}(1 - x) + h_{conv,tp} [1 + 80(x^2 - x^6)e^{-0.6Co}] \quad (5)$$

$$h_{nb} = 55 \left(\frac{p}{p_{cr}} \right)^{0.12} \left(-\log_{10} \left(\frac{p}{p_{cr}} \right) \right)^{-0.55} M^{-0.5} \left(\frac{\dot{q}_0 L_c}{A_s} \right)^{0.67} \quad (6)$$

$$h_{conv,tp} = (1 - x)h_{conv,f} + xh_{conv,g} \quad (7)$$

$$h_{conv,\phi} = \left(3.66 + \frac{0.0668 (Re_{\phi 0} Pr_{\phi} D_h / L_c)}{1 + 0.04 (Re_{\phi 0} Pr_{\phi} D_h / L_c)^{2/3}} \right) \frac{k_{\phi}}{D_h} \quad (8)$$

where h_{nb} is the nucleate boiling heat transfer coefficient, $h_{conv,tp}$ is the two-phase flow heat transfer coefficient, and $Co = [g(\rho_f - \rho_g)D_h^2/\sigma]^{-0.5}$ is the confinement number. Liquid-only (Re_{f0}) and vapor-only (Re_{g0}) Reynolds numbers, when the entire mass flux through channel occurring as liquid/vapor phase, are given as $Re_{\phi 0} = G_{ch,i} D_h / \mu_{\phi}$, where $\phi = f, g$. The heat transfer coefficient (h_{fb}) of both microchannels is calculated by taking the length-weighted-average of the heat transfer coefficient for the subcooled liquid phase, vapor-liquid mixture, and superheated vapor phase along the channel.

2.3. Eigenvalues of the dynamic system

The governing Eqs. (2) and (4) can be rewritten in dynamical form as follows:

$$\dot{X}(t) = F(X, \eta) \quad (9)$$

where $X = (G_{ch,1}, T_{w,1}, T_{w,2})^T$ are phase variables, and $\eta = (\dot{q}, C_{lat}, T_{in})^T$ are design and operating parameters. The equilibrium points of the current system can be obtained by setting the right-hand sides of the governing momentum Eq. (2) and energy Eq. (4) (or Eq. (9)) to zero such that:

$$\dot{X}(t) = 0 \Rightarrow F(X, \eta) = 0 \quad (10)$$

The equilibrium points of the 1st channel from the momentum equation are:

$$G_{ch,1}^* = \frac{G_{in}}{2} \quad (11)$$

where the asterisk denotes an equilibrium point. Therefore, the equilibrium mass flux in the second microchannel is also $G_{ch,2}^* = G_{in} - G_{ch,1}^* = G_{in}/2$. From the energy equation,

$$T_{w,i}^* = T_{f,i} + \frac{\dot{q}_0 L_c}{h_{fb,i} A_s} \quad (12)$$

Because the heat transfer coefficient and fluid temperature at the equilibrium operating conditions are equal for both parallel microchannels, it can be written $T_{w,1}^* = T_{w,2}^*$.

All these equilibrium points do not necessarily provide optimal performance, and sometimes undesirable flow maldistribution induced by the Ledinegg instability and other instabilities may occur. A general stability analysis is performed to find the stable and unstable equilibrium operating points. The stability of the equilibrium operating points ($G_{ch,1}^*$, $T_{w,1}^*$, $T_{w,2}^*$) are assessed by inspecting the eigenvalues of the Jacobian matrix (J) obtained from the governing Eqs. (2) and (4). By taking the partial derivative of Eq. (10) with respect to phase variables, the Jacobian matrix of the present system is written as follows:

$$J = \begin{bmatrix} -\frac{1}{2L_c} \left(\frac{\partial \Delta P_{ch,2}}{\partial G_{ch,1}} + \frac{\partial \Delta P_{ch,1}}{\partial G_{ch,1}} \right) & -\frac{1}{2L_c} \frac{\partial \Delta P_{ch,1}}{\partial T_{w,1}} & \frac{1}{2L_c} \frac{\partial \Delta P_{ch,2}}{\partial T_{w,2}} \\ \frac{A_s h_{fb,1}}{m_w c_w} \frac{\partial T_{f,1}}{\partial G_{ch,1}} & \frac{-h_{fb,1} A_s - C_{lat} L_c}{m_w c_w} & \frac{C_{lat} L_c}{m_w c_w} \\ \frac{A_s h_{fb,2}}{m_w c_w} \frac{\partial T_{f,2}}{\partial G_{ch,1}} & \frac{C_{lat} L_c}{m_w c_w} & \frac{-h_{fb,2} A_s - C_{lat} L_c}{m_w c_w} \end{bmatrix} \quad (13)$$

The components of the Jacobian matrix depend on many system and operating parameters, including the lateral thermal conductance and heat input to fluid. To simplify the terms for subsequent presentation, J is rewritten as:

$$J = \begin{bmatrix} a & -b & b \\ -c & d & e \\ c & e & d \end{bmatrix} \quad (14)$$

where a , b , c , d , and e are defined by the matrix entries in Eq. (13), for example, $e = \frac{C_{lat} L_c}{m_w c_w}$. The eigenvalues of the Jacobian are obtained by solving the characteristic equation,

$$|J - \lambda I| = 0 \quad (15)$$

where I is the identity matrix and the Eq. (15) is rewritten as follows:

$$\begin{vmatrix} a - \lambda & -b & b \\ -c & d - \lambda & e \\ c & e & d - \lambda \end{vmatrix} = 0 \quad (16)$$

So, the characteristic polynomial obtained by solving the above determinant is,

$$\lambda^3 - (a + 2d)\lambda^2 - (2bc - 2ad - d^2 + e^2)\lambda + 2bcd - ad^2 + 2bce + ae^2 = 0 \quad (17)$$

and the calculated eigenvalues λ from this Eq. (17) are given below as:

$$\lambda_1 = d + e \quad (18a)$$

$$\lambda_2 = \frac{1}{2} \left(a + (d - e) - \sqrt{(a - d + e)^2 + 8bc} \right) \quad (18b)$$

$$\lambda_3 = \frac{1}{2} \left(a + (d - e) + \sqrt{(a - d + e)^2 + 8bc} \right) \quad (18c)$$

The derived eigenvalues are functions of the lateral thermal conductance and heat input. For instance, the terms a , b , and c depends on the heat input, the term d depends on the heat input and lateral conductance, and the term e is a function of the lateral conductance only. The eigenvalues can be inspected to analyze the effect of these parameters on the equilibrium points to demarcate operating regions with maldistribution (unstable equilibrium points), where at least one of the eigenvalues has a positive real part, versus region with uniform distribution (stable equilibrium points) where all the eigenvalues have a negative real part. Therefore, the forms of these eigenvalue expressions can confirm the nature of how thermal coupling stabilizes systems of parallel microchannels.

2.4. Transient solution of the flow distribution

The governing equations offer a dynamic model for analyzing the transient flow distribution and wall temperature as a function of the thermal coupling between parallel microchannels. The transient response of the system is obtained by numerically solving the set of ordinary differential Eqs. (2) and (4) using the MATLAB `odes15s` function, which is chosen as these equations are stiff. This function uses a variable-step, variable-order (VSVO) solver based on the numerical differentiation formulas (NDF) of orders 1 to 5 [33]. This solves the equations as initial value problems (IVP), where the initial values are

perturbed equilibrium points ($G_{ch,1}^* \pm \delta G_{ch,1}$; $T_{w,1}^* \pm \delta T_{w,1}$; $T_{w,2}^* \pm \delta T_{w,2}$).

3. Results and discussions

In these results, Section 3.1 first identifies the operating regions corresponding to uniformly distributed versus maldistributed flow, obtained by the stability analysis approach as described in Section 3.1. Next, Section 3.2 presents the transient system response for various lateral thermal conductances and heat inputs that result in different flow distributions, obtained via transient solution of the governing equations as described in Section 2.4. The dependence of the system eigenvalues on various design and operating parameters is then discussed in detail (Section 3.3). Section 3.3.1 reveals that the first and second eigenvalues do not participate in determining the flow distribution of the system. Section 3.3.2 describes the role of the third eigenvalue in governing the flow distribution as a function of thermal coupling and heat input, and a scaling analysis of the third eigenvalue in Section 3.3.3 explains the asymptotic nature of the stability boundary. Lastly, Section 3.4 explores the dynamic response for scenarios at which the stability boundary is has asymptotically converged; although maldistribution is guaranteed in microchannels for sufficiently high heat inputs, thermal coupling can nevertheless reduce the severity of the wall temperature difference in the presence of maldistribution.

3.1. Mapping the regions of flow stability and maldistribution

The eigenvalues of the parallel microchannel system depend on several operational and design parameters as governed by Eq. 16. Therefore, a stability map is generated by calculating the eigenvalues of the Jacobian matrix for varying lateral thermal conductance C_{lat} from 0–600 W/mK and heat input q' from 228–242 W/m. All other design and operating parameters are kept constant, as given in Table 1. Stable regions will have uniform flow distribution and unstable regions will have flow maldistribution, assessed per the sign of the positive real parts of the eigenvalues as discussed in Section 2.3.

Fig. 3 displays a stability map for the parallel microchannels in the parametric plane of the lateral thermal conductance C_{lat} and heat input q' , where blue marked points indicate stable flow (no marks are shown

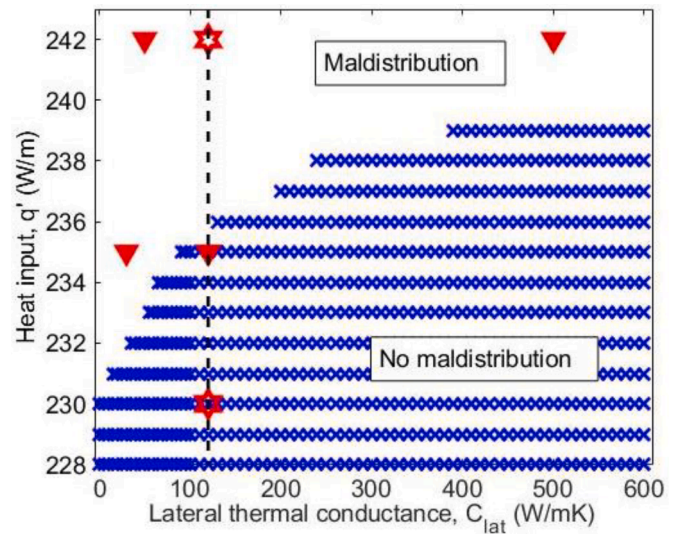


Fig. 3. Stability map showing over the parametric plane of lateral thermal conductance (C_{lat}) and heat input (q') to the system of two thermally coupled parallel channels. The map demarcates the unstable (maldistribution) and stable (no maldistribution) regions, where the stable region is indicated by blue marks. Filled red triangle symbols indicate specific operating points at which the transient flow distribution dynamics are discussed in Sections 3.2 and 3.4.

for unstable flow). The entire plane is divided by a stability boundary between maldistributed (unstable) and uniformly distributed (stable) flow. It is evident from the figure that the flow is always stable for low heat inputs $\dot{q}' < 230$ W/m, below the threshold at which the liquid reaches the saturation temperature at the outlet of microchannels for uniform flow distribution. This is because, as reviewed in Section 1.1, the Ledinegg instability mechanism only occurs in the intermediate mass flux region where there is a vapor-liquid mixture at the channel outlet. Without any lateral thermal conductance ($C_{lat} \approx 0$), the flow becomes maldistributed for any heat input $\dot{q}' > 230$ W/m. The stability of microchannels improves with increasing non-zero thermal coupling. Hence, the stability boundary increases to higher heat inputs with increasing lateral thermal conductance, as observed in Fig. 3. The trend in the stability boundary with increasing thermal coupling depicts a few distinct characteristics. Initially, there is a steep increase in the stability boundary from zero to low lateral thermal conductance. However, this stability enhancement asymptotically converges to a particular heat input at very high lateral thermal conductance, indicating that no level of thermal coupling can avoid maldistribution above this heat input. The same asymptotic maldistribution stability boundary has been previously observed (but not explained) via the finite volume simulations of Van Oevelen et al. [30], confirming that the currently developed lumped model can accurately capture the effect of thermal coupling on flow maldistribution.

The enhancement of stability with thermal coupling can be generally explained as follows. In the absence of thermal coupling, heat transfer from one channel to the other is prohibited; any perturbation in mass flux unstably grows due to the negative slope of the single-channel load curve (i.e., a slight reduction in flow rate to one channel increases the pressure drop and causes a further flow rate reduction in an excursive manner). This leads to a highly maldistributed flow where one channel gets much less mass flux than the other to equalize the pressure across both channels. As a result, the wall temperature in the flow-starved channel is also becomes higher than the other channel. However, with the mechanism of thermal coupling enabled, this increasing wall temperature of the flow-starved channel causes heat transfer to the other channel via thermal conduction between the two channels. With sufficient thermal conductance, this allows the flow-starved channel to return to back its initial condition after releasing the heat stored in the channel wall. While this improvement to stability is simply explained, it is not trivial to understand why the improvement to stability asymptotes with increasing thermal coupling. The modeling approach implemented in the current work offers a mechanistic explanation of this trend in the stability boundary, as will be discussed in the upcoming sections.

3.2. Dynamic system response for stable versus unstable conditions

The previous Section 3.1 introduced the stability map for the present system having two distinct regions associated with uniformly distributed and maldistributed flow in the parametric plane of lateral thermal conductance (C_{lat}) versus applied heat input ($\dot{q}'$). However, this map can only predict the occurrence of maldistribution via stability analysis, whereas in this Section 3.3, the transient governing equations are solved to analyze the flow maldistribution dynamics and its effect on the wall temperature.

The two operating points for which these transient simulations are performed lie on either side of the stability boundary for lateral thermal conductances of 30 and 120 W/mK at $\dot{q}' = 235$ W/m. The rest of the design and operating parameters are as specified in Table 1. The lower lateral conductance corresponds to a point in the unstable region, while the higher is in the stable region, both marked as red filled triangles in Fig. 3.

Fig. 4 shows the temporal behavior of the mass flux and wall temperature in the parallel microchannels, with flow maldistribution (Fig. 4(a and b)) and without flow maldistribution (Fig. 4(c and d)). Fig. 4(a)

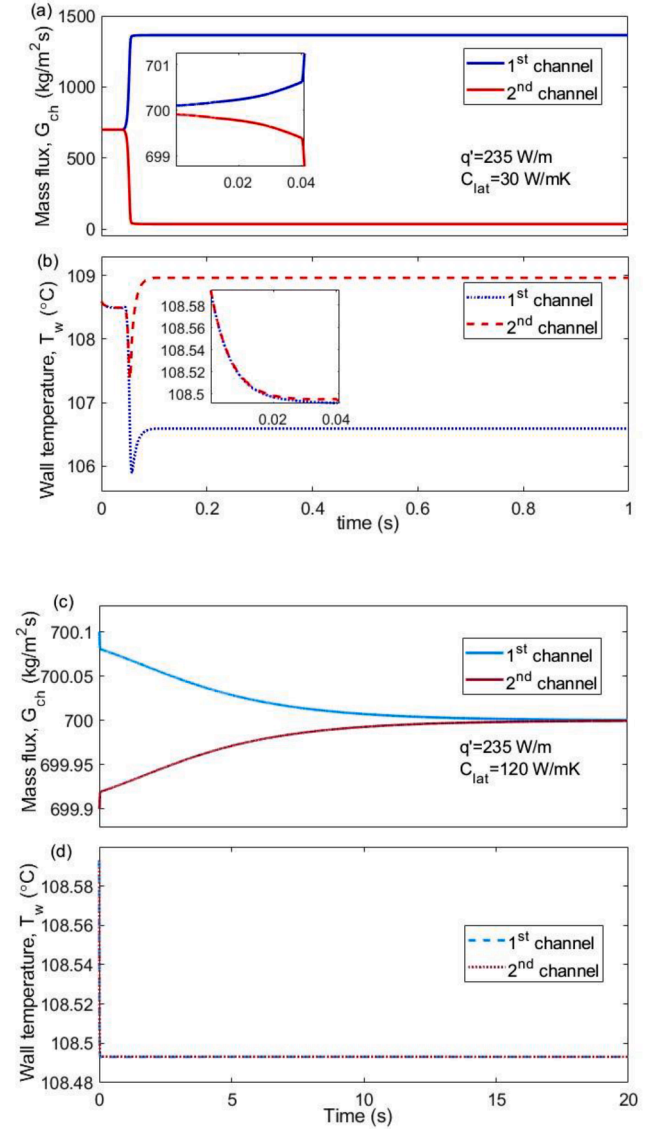


Fig. 4. Transient (a,c) mass flux and (b,d) wall temperature for flow through two parallel microchannels at $\dot{q}' = 235$ W/m, where different line styles (solid versus dotted) are used to distinguish between the response of each channel. (a, b) corresponds to a case with maldistribution ($C_{lat} = 30$ W/mK) whereas (c,d) corresponds to a case without maldistribution at a higher lateral thermal conductance ($C_{lat} = 120$ W/mK).

shows the transient mass flux with the occurrence of maldistribution for a low lateral thermal conductance. With an external disturbance to the steady state operating point, the mass flux in both channels diverges in opposite directions from the initially uniform mass flux $G_{ch,1} = G_{ch,2} = 700$ kg/m²s (as shown in the zoomed inset). Once the system has stabilized, one microchannel is flooded with nearly all of the available mass flux $G_{ch,1} = 1380$ kg/m²s, while the other is starved of fluid, $G_{ch,2} = 2$ g/m²s. These two new stable operating points correspond to P_1 and P_2 as discussed in introduction section, Fig. 1. As a result, flow through the microchannels remains maldistributed, which affects the wall temperatures as plotted in Fig. 4(b). The uniform initial mass flux means that both microchannels initially have the same wall temperature. The wall temperatures start to deviate from this condition as soon as the mass fluxes become non-uniform. The channel receiving the lower mass flux has vapor at the channel outlet, corresponding to a deteriorated heat transfer coefficient and increased wall temperature. The difference in the wall temperatures reaches a maximum when uneven mass flux first

reaches its absolute peak, and eventually settles to a fixed difference.

Fig. 4(c and d) exhibits the flow distribution and wall temperature of the microchannels for a higher lateral thermal conductance in a stable region (no maldistribution). Following a small flow disturbance, the mass flux through the microchannels slowly converge with time till both have the same uniform flow. As described above at the end of Section 3.1, this stability occurs because when one microchannel has a lower mass flux and the wall temperature tries to increase, i.e., $T_{w,1}(t) > T_{w,1}^*$; the increased thermal coupling enables sufficient heat transfer between the microchannels such that the system can return to the uniform distribution. This process of heat transfer prevents the microchannels from developing flow maldistribution, whereas in the case of the lower lateral thermal conductance $C_{lat} = 30$ W/mK, the heat transfer between microchannels is insufficient to mitigate the flow maldistribution.

3.3. Dependence of eigenvalues on parameters

The stability map of flow maldistribution shown in Fig. 3 is based on evaluation of the eigenvalues of the Jacobian matrix for the present system. If at least one eigenvalue has a positive real part, the flow is unstable. With the effect of lateral thermal coupling on flow maldistribution established above, this section explains these trends based on the dependency of the eigenvalues on the heat input and lateral thermal conductance.

Fig. 5 plots representative eigenvalues of the Jacobian matrix calculated as a function of the heat input in the presence of a given lateral thermal conductance ($C_{lat} = 120$ W/mK). The vertical axis indicates the value of the real part of the eigenvalues, with emphasis on whether these are positive or negative with respect to the solid black horizontal zero line. It is observed from the figure that for lower heat inputs $q' < 230$ W/m for which the subcooled liquid entering the channel does not reach the saturation temperature at the outlet, all three eigenvalues have negative real parts. So, for these powers, the uniform flow distribution inside the microchannels is stable. As the heat input applied to the microchannels further increases to $q' > 230$ W/m, the third eigenvalue, λ_3 , (red dotted line) starts to increase toward the positive side of the axis. This eigenvalue λ_3 remains negative between 230 W/m $< q' \lesssim 236$ W/m, and hence there is no maldistribution despite flow boiling in the channels owing to the thermal coupling. However, for a high heat input $q' > 236$ W/m, this eigenvalue becomes positive and maldistribution appears. The star points on the curve of λ_3 correspond to points on a vertical dashed line of Fig. 4 where these points are stable (no

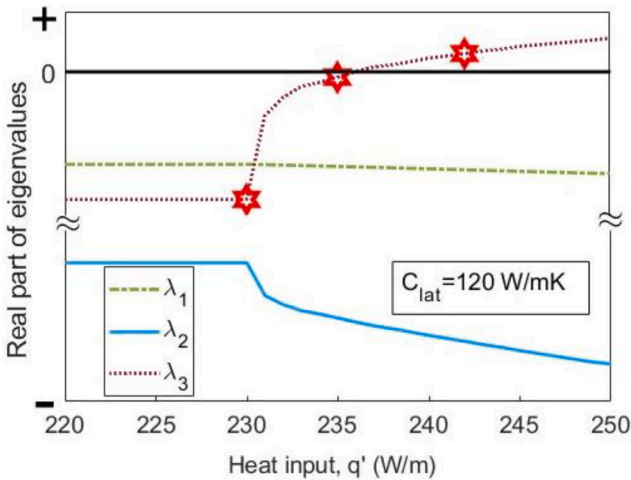


Fig. 5. Dependency of the system eigenvalues λ_1 , λ_2 , and λ_3 with respect to the heat input q' for a given lateral thermal conductance ($C_{lat} = 120$ W/mK). The first and second eigenvalues λ_1 and λ_2 remain negative for all heat inputs whereas the third eigenvalue λ_3 becomes positive at some particular heat input.

maldistribution) at $q' = 230$ W/m, near to the stability boundary (but on the stable side) at $q' = 235$ W/m, and unstable (maldistribution) at $q' = 242$ W/m. Alternatively, the first and second eigenvalues of the system (λ_1 and λ_2) always lie on the negative side of the axis for the entire range of heat input.

3.3.1. Scaling analysis of the first and second eigenvalues

It was observed in Fig. 5 that the real parts of the first and second eigenvalues (λ_1 and λ_2) were always negative (stable) for a specific value of lateral thermal conductance $C_{lat} = 120$ W/mK. A scaling analysis in this subsection demonstrates that these two eigenvalues will remain negative for all possible values of heat input and lateral thermal conductance.

The first eigenvalue is written here again

$$\lambda_1 = d + e = \frac{-h_{fb,1}A_s - C_{lat}L_c}{m_w c_w} + \frac{C_{lat}L_c}{m_w c_w} = \frac{-h_{fb,1}A_s}{m_w c_w} < 0 \quad (19)$$

where the simplified expression only contains the convective heat transfer coefficient $h_{fb,1}$, the heat transfer area of the microchannel A_s , and the specific thermal capacity of the heat sink $m_w c_w$. All these quantities are universally positive, and as a result, λ_1 will always remain negative. Furthermore, it is independent of lateral thermal coupling.

The second eigenvalue λ_2 is a non-trivial function of a , b , c , d and e , the components of J as defined by Eqs. (14) and (15):

$$\lambda_2 = \frac{1}{2} \left(a + (d - e) - \sqrt{(a - d + e)^2 + 8bc} \right) \quad (20)$$

If the term outside of the squared root is less than the squared root term for any heat input and lateral thermal coupling, then the real part of λ_2 is negative. We first inspect the difference between d and e as follows:

$$(d - e) = \frac{-h_{fb,1}A_s - C_{lat}L_c}{m_w c_w} - \frac{C_{lat}L_c}{m_w c_w} = \left(\lambda_1 - \frac{2C_{lat}L_c}{m_w c_w} \right) < 0 \quad (21)$$

It is established that λ_1 is always negative, and the thermal conductance C_{lat} and microchannel length L_c cannot be negative, so $(d - e)$ is always negative. Therefore, the sign of λ_2 depends only the comparative values of a versus $\sqrt{(a - d + e)^2 + 8bc}$. The term $(a - d + e)^2 = \{a - (d - e)\}^2$ is always positive and greater than a by inspection because $(d - e) < 0$. Additionally, the term $8bc$ is always greater than zero because the partial derivative of channel pressure drop with respect to wall temperature $b = (\partial \Delta P_{ch,2} / \partial T_{w,2}) / 2L_c$ is positive, and $c = -A_s h_{fb,1} (\partial T_{f,1} / \partial G_{ch,1}) / m_w c_w$ is also positive. As a result, this confirms that $\sqrt{(a - d + e)^2 + 8bc} > a$ and therefore $\lambda_2 < 0$. This scaling analysis confirms that the second eigenvalue λ_2 always remains negative and does not cause flow maldistribution in the parallel microchannel system. By deduction, the third eigenvalue λ_3 solely cause flow maldistribution in the current system.

3.3.2. Dependence of the third eigenvalue on lateral thermal coupling

This section analyzes the dependency of the third eigenvalue λ_3 on the magnitude of heat input and lateral thermal coupling. Fig. 6 plots λ_3 as a function of heat input for multiple different increasing values of the lateral thermal conductance $5 < C_{lat} < 500$ W/mK. As described earlier in Fig. 6, λ_3 is negative below $q' = 230$ W/m for which there is subcooled liquid exiting the channel. For $q' > 230$ W/m, λ_3 immediately crosses from negative to positive and the system is unstable if the lateral thermal coupling is zero or negligible. With increasingly non-zero lateral thermal coupling, this crossover from negative to positive is delayed to higher heat inputs. For example, the eigenvalue λ_3 remains negative for a small additional range of heat input $230 \lesssim q' < \sim 231$ W/m for a lateral thermal conductance $C_{lat} = 10$ W/mK. Whereas a larger lateral thermal conductance $C_{lat} = 100$ W/mK offers a wider range of stability for heat

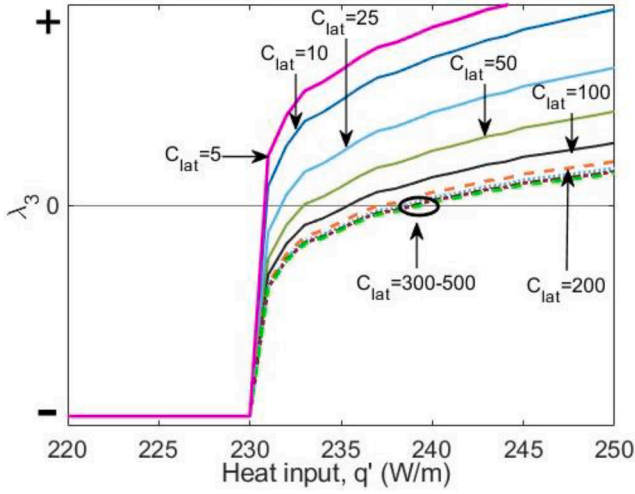


Fig. 6. Functional dependence of the third eigenvalue λ_3 for varying lateral thermal conductance and heat input. There is a shift in the heat input at which the value becomes positive for increasing C_{lat} from 0 to ≈ 200 W/mK; however, further increases in $C_{lat} \gtrsim 300$ –500 W/mK no longer affect λ_3 .

inputs $230 \lesssim q' \lesssim 235$ W/m. These λ_3 values confirm that higher thermal coupling has a more substantial effect on mitigating maldistribution. However, the improvement to flow stability comes to an end after the λ_3 curve converges to a single line that crosses from negative to positive at a fixed heat input of $q' = 240$ W/m for very large lateral thermal conductance (in the present case $C_{lat} \gtrsim 300$ –500 W/mK). This converged λ_3 profile is consistent with the asymptotic stability boundary as shown in Fig. 3. A scaling analysis of λ_3 in the next Section 3.3.3 explains this convergent behavior.

3.3.3. Scaling analysis of the third eigenvalue to explain the asymptotic stability boundary

This section describes the underlying reason why the stability boundary for flow boiling in parallel microchannels is asymptotic with increasing lateral thermal coupling, i.e., that there is a limit to mitigating maldistribution, via scaling analysis of the third eigenvalue λ_3 in the limit of high thermal conductance. The third eigenvalue is written below:

$$\lambda_3 = \frac{1}{2} \left(a + (d - e) + \sqrt{(a - d + e)^2 + 8bc} \right) \quad (22)$$

The components $d = (-h_{f1}A_s - C_{lat}L_c)/m_w c_w$ and $e = C_{lat}L_c/m_w c_w$ are proportional functions of the thermal coupling coefficient; the term $(d - e)$ is always negative and becomes increasingly negative with thermal coupling. Hence, in the limit of high C_{lat} , $(a - d + e)^2 = (a - (d - e))^2$ becomes significantly larger than $8bc$ which only depends on thermodynamic properties via b and c , and therefore $\sqrt{(a - d + e)^2 + 8bc} \approx a - d + e$.

After considering these scaling assumptions, in the limit of $C_{lat} \rightarrow \infty$, the third eigenvalue reduces to simply:

$$\lambda_3 \approx a \quad (23)$$

It hence only depends on the Jacobian element $a = -(\partial \Delta P_{ch,1} / \partial G_{ch,1} + \partial \Delta P_{ch,2} / \partial G_{ch,2}) / 2L_c$ defining the slope of microchannel pressure drop depending on the thermodynamic properties calculated based on heat input to microchannels, regardless of the thermal coupling coefficient. Therefore, λ_3 becomes dictated by only the fluid thermophysical properties at sufficiently high lateral thermal coupling. It is confirmed via stability and scaling analysis using our novel lumped model that fluid properties ultimately govern the stability behavior in the limit of high lateral thermal coupling, posing a limit to

which lateral thermal coupling can altogether avoid flow maldistribution. This explains the mechanism behind the asymptotic nature of the flow maldistribution stability boundary with increasing lateral thermal coupling.

3.4. Intensity of thermal coupling coefficient in maldistribution region

It is established in the previous sections that only the third eigenvalue (λ_3) participates in causing the flow maldistribution. Moreover, scaling analysis of this eigenvalue explains why the impact of lateral thermal coupling reaches a limit in preventing maldistribution. Although the linear stability behavior is limited, this section reveals that lateral thermal coupling can nevertheless reduce the consequences of flow maldistribution, even when it cannot be avoided altogether.

Here we analyze the flow dynamics in the unstable region at applied heat input of $q' = 242$ W/m and lateral thermal conductances of $C_{lat} = 50$ W/mK and $C_{lat} = 500$ W/mK corresponding to the filled triangle in Fig. 3, with all other quantities as specified in Table 1. Fig. 7 plots the transient (a) mass flux and (b) wall temperature for the above conditions when subjected to a perturbation from the uniformly distributed flow state. Both higher lateral thermal conductance ($C_{lat} = 500$ W/mK) and smaller lateral thermal conductance ($C_{lat} = 50$ W/mK) cases eventually exhibit flow maldistribution as expected based on the linear stability analysis as shown in Fig. 7(a). However, the heat sink wall temperature dynamics are still affected by lateral thermal coupling, as shown in Fig. 7

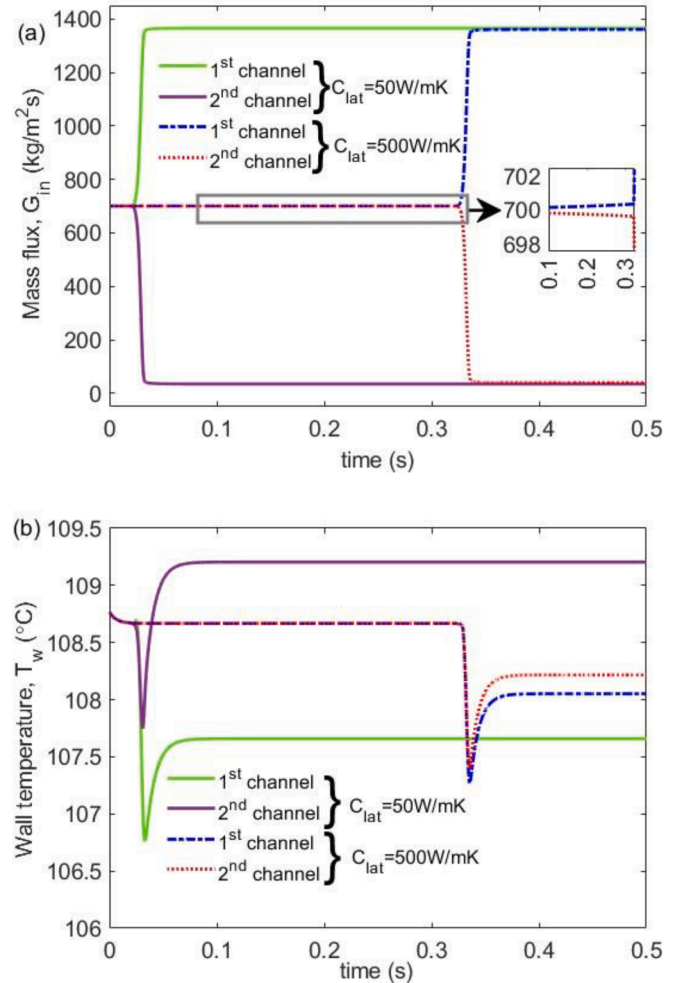


Fig. 7. Effect of lateral thermal coupling on (a) mass flux maldistribution and (b) wall temperature difference for both high (dotted lines; $C_{lat} = 500$ W/mK) and low (solid lines; $C_{lat} = 50$ W/mK) lateral thermal conductance.

(b). Prior to maldistribution, the wall temperatures are equal for both microchannels. In both cases wall temperature difference between microchannels appears as soon as flow maldistribution occurs. However, the temperature difference between microchannels is more severe for lesser thermal coupling, leading to a higher maximum temperature in the flow-starved channel. The lesser temperature difference between the channels for the higher lateral thermal coupling can be attributed to lateral heat transfer. Therefore, although the parallel system is linearly unstable, causing flow maldistribution, the consequences such as risk of thermal runaway or thermal stressed are nevertheless reduced.

4. Conclusions

Flow maldistribution in parallel boiling microchannels, and the effect of lateral thermal coupling, are analyzed using a lumped flow model. A stability analysis performed using the developed model reasserts the influence of lateral thermal coupling on flow maldistribution found in earlier studies using more complex numerical flow models. Namely, without thermal coupling, flow maldistribution occurs in parallel microchannels for any heat input that causes the subcooled inlet liquid to boil before exiting the channel. With thermal coupling, flow maldistribution can be avoided up to higher heat input thresholds. However, in the limit of very large thermal coupling, it no longer provides a benefit in terms of linear stability, and there is a critical heat input above which the flow maldistribution occurs regardless of thermal coupling.

This asymptotic nature of the maldistribution stability boundary is explained for the first time in this work, owing to the ability to assess the dependency of eigenvalues on thermal coupling using our lumped model. A scaling analysis reveals that two of the eigenvalues are not involved in flow maldistribution. Therefore, the occurrence of flow maldistribution is defined solely by the third eigenvalue. The third eigenvalue is a nonlinear function of the thermal coupling and fluid properties. Scaling analysis shows that in the limit of very high thermal coupling, this eigenvalue is solely dependent on the thermodynamic properties of the fluid. Therefore, the third eigenvalue changes nonlinearly for low thermal coupling, but converges to a single stability threshold in the limit of high coupling, providing the reasoning for the asymptotic shape of the stability boundary.

Transient solutions of the lumped models are performed to illustrate the flow and wall temperature dynamics for stable and unstable cases. In the unstable region, the maldistribution between the microchannels is extreme; as a result, the flow-starved microchannel has a higher wall temperature. Although the linear stability confirms that very high lateral coupling cannot entirely avoid maldistribution, the dynamic solutions reveal that there is nevertheless a lesser difference between the wall temperatures of each microchannels in the presence of maldistribution, thereby reducing the consequences of maldistribution.

CRedit authorship contribution statement

Md Emadur Rahman: Writing – original draft, Methodology, Investigation, Conceptualization. **Justin A. Weibel:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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